

43030 Professional Practice in Computing**Assessment Task 3: Professional Learning Report**

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Portfolio: <https://bellelynn.github.io/uts-43030-portfolio/>**1. Introduction**

This report reflects on the professional development I have undertaken in fulfilment of the learning plan I submitted in Assessment Task 1. The plan committed roughly 150 effort hours across three learning goals, each chosen to close a specific gap between where I am now and what my target employers — Holocentric and Fire Front — actually ask for in their graduate job ads. Both build regulated Australian software, which means engineers there need to ship full-stack code and defend their privacy, security, and ethics decisions in the same week.

The structure of this report mirrors the original PLP. Section 2 explains why I chose each activity, Section 3 presents what was produced, Section 4 links this report back to my portfolio and the original plan via an evidence index, Section 5 is the reflection on what changed in my thinking, and Section 6 sets out three concrete future goals. Together these sections demonstrate progress against the three graduate attributes the subject is built around — socially responsible, technically proficient, and collaborative and communicative (Australian Computer Society [ACS], 2022). All supporting artefacts — code, audit documents, course progress, and demo videos — are hosted in the public portfolio linked on the title page.

2. Justification of Professional Development Activities

My initial skills audit pointed at three priority gaps: (i) a legalistic rather than architectural understanding of privacy and professional ethics; (ii) limited exposure to defensive cybersecurity, particularly at the network layer; and (iii) no real experience with the JavaScript ecosystem (modern front-end frameworks plus a Node-based back-end), which is the dominant stack in every full-stack graduate ad I shortlisted from Holocentric and Fire Front. Each activity below targets one of those gaps.

2.1 Activity 1 — Ethical principles and privacy compliance

Holocentric develops enterprise Risk and Compliance platforms for Australian government and regulated industries. Reading through their case studies and recent job ads showed me that engineers there are expected to treat privacy as a first-class architectural concern, not a post-hoc legal review. My own situation was the opposite — I could quote the Australian Privacy Principles, but I had never let them change a database schema. So I committed to two things in parallel: finishing the UTS Canvas modules on professional ethics, the IT code of ethics, and Indigenous data sovereignty, and using what I learned to audit the data model of LEXIS, my Law Firm Case Management System. Pairing the theory with an applied audit was deliberate — theory on its own would not have changed how I write code.

2.2 Activity 2 — Foundational cybersecurity skills

My original plan scoped this goal around the OWASP Top 10, which is all application-layer. Three weeks in, I changed my mind. Two informational interviews with UTS alumni in security-adjacent roles, plus a closer reading of the Australian Cyber Security Centre's Essential Eight (ACSC, 2023), made it clear that without a network-layer foundation I could not really reason about the threat model my code runs inside. I changed direction and enrolled in the Cisco Networking Academy's Network Security course instead,

treating it as a base I could layer application-layer security on top of later. Rescoping mid-semester was uncomfortable at the time, but being able to explain why I did it is itself the kind of strategic self-management the subject treats as a graduate attribute.

2.3 Activity 3 — Full-stack development on the JavaScript ecosystem

Every Holocentric and Fire Front graduate ad I shortlisted assumed familiarity with JavaScript across the stack — a modern front-end framework (React, Angular, or Vue) and, increasingly, a Node-based back-end as well. My prior experience was solid on C# and ASP.NET Core, but that left me with a narrow, single-language profile that did not match what the actual market was hiring for. So I committed the largest single block of effort this semester (around 70 hours) to learning the JavaScript ecosystem end-to-end and building LEXIS, a full-stack case management application, entirely on it: React with Vite on the front-end, Node.js with Express on the back-end, and PostgreSQL underneath. Building a non-trivial application across the whole stack — not following a tutorial — was deliberate. It forced me to deal with the real integration problems (CORS, async state, transaction-level data integrity, error propagation across the network boundary) that a single-language project would have hidden from me.

3. Evidence of Achievement

This section summarises what each activity produced. All artefacts — completion screenshots, audit documents, code commits, and the full LEXIS source repository — are hosted in the public portfolio linked on the title page, under the three sections referenced in Section 4.

3.1 Ethical principles and privacy compliance

I completed all the required UTS Canvas modules covering professional ethical principles, the ACS Code of Professional Conduct, the Australian Privacy Principles (Office of the Australian Information Commissioner [OAIC], 2022), and Indigenous data sovereignty (Maiaam nayri Wingara Indigenous Data Sovereignty Collective, 2018). Beyond ticking those off, I produced two concrete deliverables. First, I ran a privacy audit on the LEXIS PostgreSQL schema, classifying every personal-data field into three categories (client identifiers, matter details, billing data) and documenting the legal basis, retention period, and authorised role for each. Second, I refactored the data access layer so role-based access control (RBAC) is enforced inside the SQL queries — every read against client or case data now includes a tenant- and role-scoped WHERE clause, rather than relying on the route handler alone. The original design quietly allowed a class of accidental data exposure; the refactored one does not.

3.2 Foundational cybersecurity

I completed the majority of the Cisco Network Security course, including the modules on secure network design, access control, firewall technologies, and IPS. I deliberately did not sit the final certification exam — the practical lab content was worth doing properly, and partial completion with real understanding was a more honest outcome than a finished certificate I rushed through. The course has already changed how I write code: I now reason explicitly about which interfaces of the LEXIS API are network-exposed, and I have added defensive middleware to the Express back-end — body-size limits, content-type validation, JWT verification at the route boundary, and per-IP rate limiting — as a direct result.

3.3 Full-stack development on the JavaScript ecosystem

I shipped LEXIS as a complete single-page web application. The front-end uses React 19 with Vite as the build tool, Ant Design 6 for the component library, ECharts for the dashboard's case-growth chart, React

Router DOM v7 for client-side routing, and Axios for HTTP. The back-end is a Node.js / Express 5 REST API talking to PostgreSQL via the pg driver, with auth handled by JWT bearer tokens issued at login. The database is fully normalised across five tables (Office, Lawyer, Client, Case, Case Assignment) with foreign-key constraints and a many-to-many junction table for case-lawyer assignments. All four core workflows scoped in the PLP are working end-to-end: case listing with sort and filter, case detail with reassignment, client management, and a statistics dashboard. The complete source code is public at github.com/bellelynn/case-management.

4. Integration with the PLP and Portfolio

This Report, my portfolio, and the original PLP are meant to be read as one consistent narrative. The PLP set the three goals and the success criteria. The portfolio holds the underlying artefacts — the audit document, the RBAC refactor, the Cisco progress evidence, and the full LEXIS source repository — organised under one section per PLP goal, so any claim made in this Report can be traced directly to its primary evidence. The table below provides the index.

PLP Goal	Portfolio Section	Key Artefacts
Goal 1 — Ethics & Privacy	/ethics-privacy/	Canvas module screenshots; LEXIS privacy audit; RBAC enforcement in SQL queries; data classification table
Goal 2 — Cybersecurity	/cybersecurity/	Cisco Networking Academy progress dashboard; module-level completion screenshots; LEXIS Express middleware (body-size, content-type, rate-limit); mid-semester rescope note
Goal 3 — Full-Stack (LEXIS)	/lexis/	Full source code at github.com/bellelynn/case-management ; architecture diagram; LEXIS UI screenshots; CaseList component code

Two things shifted from the original plan and I want to flag them rather than hide them. The cybersecurity goal was rescope mid-semester from application-layer to network-layer foundations, as justified in Section 2.2; the rescope note in the portfolio's cybersecurity section keeps that change on the record. The privacy goal also grew — the audit surfaced an opportunity to push RBAC enforcement into the SQL layer, and that extra work sits under the same PLP goal in the portfolio rather than being broken out separately, which keeps the one-to-one mapping between PLP goals and portfolio sections intact.

5. Reflection on Progress and Learning

Schön's (1983) distinction between reflection-in-action and reflection-on-action gave me a useful frame for this section. Rather than retelling what happened, I have tried to be honest about what I now understand differently because of it.

5.1 From legal theory to architectural decisions

The biggest shift came from the ethics work. Before this semester, I treated privacy as a checklist at the end of a project — write a notice, tick a box, move on. Auditing my own database forced me to see that every schema decision is already a privacy decision: what gets collected, how long it is kept, who can join what to what. The Indigenous data sovereignty module pushed this further by challenging an assumption I had not noticed I was making — that data, once you have collected it, simply belongs to you. I now

design data models by answering two questions up front: who has authority over this data, and what is the minimum collection that does the job? That is a real change in how I work, not a cosmetic one.

5.2 Replanning is a professional skill, not a failure

Rescoping cybersecurity, and deciding not to chase the Cisco certification at the cost of understanding, were both uncomfortable calls at the time. Looking back, making them — and being able to justify them on paper — is itself the kind of thing employers actually want. Honest progress against a revised plan is worth more than nominal completion of an unrealistic one. I expect to hit the same trade-off in industry, where renegotiating scope without losing trust is just part of the job.

5.3 The real cost of switching language ecosystems

Moving from C# to the JavaScript stack was the hardest part of the semester. It was not just learning React — it was unlearning the synchronous, strongly-typed, batteries-included mindset that ASP.NET Core had given me. Express does almost nothing for you; the JavaScript event loop turns every database call into an opportunity for a subtle async bug; and React's declarative model required me to throw out object-lifecycle intuitions that had felt obvious for years. The single worst offender was the `useEffect` dependency array — my instinct from C# component lifecycles was systematically wrong, and I burned a few evenings debugging re-render loops before my mental model caught up. The takeaway is bigger than React or Express: real ecosystem fluency is not vocabulary, it is a rearrangement of how I think about state, time, and failure. Whatever stack I learn next will need the same humility.

5.4 What I would do differently

Two things. I would start the JavaScript stack earlier — I underestimated how much head-space the ecosystem shift would eat, and I had less time than I wanted for integration polish at the end. And I would track hours against goals, not just against the calendar. The mid-semester rescoping of cybersecurity could have happened two weeks sooner if I had been watching where my effort was actually going.

6. Future Professional Development Goals

The three goals below are sequenced on purpose. Each one builds on capability established this semester, and each is scoped tightly enough to be assessable rather than aspirational.

6.1 Goal 1 — Cloud deployment proficiency (AWS or Microsoft Azure)

Rationale: LEXIS currently runs on localhost. Holocentric and Fire Front deploy to Azure and AWS respectively, and cloud literacy is a recurring requirement in graduate role descriptions. Without it, the full-stack work completed this semester cannot really be demonstrated to anyone outside a screen-share. Plan: within the next three months I will complete the AWS Certified Cloud Practitioner certification (estimated 40 hours of study) and, in parallel, deploy LEXIS to AWS using a containerised pipeline (Docker, ECS Fargate, and RDS for PostgreSQL). Success will be evidenced by a public URL, an architecture diagram, and a passing exam score, all linked from the portfolio.

6.2 Goal 2 — Application-layer security and secure development lifecycle

Rationale: this semester's cybersecurity work deliberately stopped at the network layer. The complementary application-layer knowledge — the OWASP Top 10, threat modelling, dependency scanning — is the obvious next step, and it links the cybersecurity track directly back to the LEXIS codebase I just built (OWASP Foundation, 2021). Plan: over the next four months I will (i) work through

the OWASP WebGoat exercises against a local instance, (ii) perform a documented STRIDE threat-modelling exercise on LEXIS, and (iii) integrate a dependency-vulnerability scanner (npm audit, Dependabot, or Snyk) into the project's CI pipeline. The deliverable will be a written threat model and a green CI badge on the public repository.

6.3 Goal 3 — Responsible integration of AI APIs into production workflows

Rationale: generative AI now shows up in most Australian software job ads, and Holocentric's recent product roadmap explicitly mentions AI-assisted compliance review. Being able to show I can integrate an LLM responsibly — paying attention to cost, prompt injection, and data handling — is a real differentiator at graduate level. Plan: within six months I will design and ship a document-summarisation feature inside LEXIS, calling either the Anthropic or OpenAI API from the Express back-end. The implementation will include a clear data-handling boundary (no client-identifying information leaves the system without explicit user action), structured logging of API costs, and basic prompt-injection mitigations. That closes the loop on Goal 1 from this semester: ethics stays a first-class architectural concern, not a postscript.

7. Conclusion

Across the three planned activities I moved from a legalistic to an architectural understanding of privacy, from no defensive-security background to a working network-layer foundation, and from a single-language C# profile to a JavaScript-ecosystem full-stack capability with LEXIS as the demonstrable output. More importantly, the experience of replanning mid-semester, choosing comprehension over a quick certification, and rebuilding my mental model under React and Express has changed how I expect to learn at work. The three future goals above are scoped to be completed in the six months after this assessment, and they are aimed directly at the graduate full-stack engineering roles I have targeted from the start.

8. References

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